

Founder control and breakthrough innovation: Evidence from high-tech firms in China[☆]

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ABSTRACT

We examine the effect of founder control on breakthrough innovation among Chinese high-tech firms from 2008 to 2022. We measure founder control by the equity held by founders actively involved in governance and identify breakthrough innovation with Sentence-BERT (SBERT) embeddings of patent texts. We find a strong positive association between founder control and future breakthrough innovation. Using unexpected founder departures as an instrument, our identification strategy establishes a robust causal relationship. The underlying mechanisms include talent investment, internal discipline, strategic reorientation, and technological resilience. The effect is heterogeneous across founding team's size, internal structure, attributes, and tenure. Our results support a structural view of founder governance where concentrated control serves as an engine for corporate innovation.

1. Introduction

Founder-led governance, a structure that concentrates power to shape high-risk, long-horizon innovation, has become a prominent feature of the global technology sector. While prevalent in Silicon Valley, such a structure has been conspicuously absent in mainland China. The 2024 Company Law Amendment in China represents a major liberalization of the longstanding "one-share-one-vote" principle by institutionalizing the dual-class share (DCS) structure nationwide, granting founders disproportionate voting rights. Its central premise is that enhanced voting rights can insulate visionary founders from short-term market pressures, thereby enabling the strategic autonomy required for transformative projects. Yet this insulation also raises fundamental questions of accountability. Concentrated control may just as easily entrench founders, allowing them to prioritize private benefits over long-term value creation, particularly in emerging markets where external governance is still maturing. This institutional reform thus creates a compelling setting to address a central question: *Does founder control ultimately foster or constrain breakthrough innovation?*

The literature on corporate governance reveals a central tension regarding founder control. According to conventional theory, founders should relinquish control to professional management as firms scale (Chang & Shim, 2015). While this conventional wisdom may hold for optimizing stable operations, it is insufficient in innovation-intensive contexts where a growing body of evidence suggests powerful founders can be critical for innovation performance (Colombo & Grilli, 2010; Osses et al., 2025). This scholarly divide

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persists, with some studies documenting the liabilities of founder CEOs (Zhong et al., 2023) while others highlight the unique advantages that founders provide (Kannan-Narasimhan et al., 2023). We argue that this contradiction is better understood as a product of a narrow focus on the founder as an actor. Past research has often tended to conflate the founder's transient characteristics with the enduring power of the governance structure they command. Much of this debate remains anchored in positional or biographical attributes, such as whether the founder serves as CEO (Pan et al., 2024), holds dual roles (Li et al., 2024), or possesses specific traits (Harrison et al., 2024). What continues to be underexplored is not whether a founder is present, but how the concentration of control itself functions as a structural driver for technological choice under ambiguity. Advancing this debate, therefore, requires moving beyond questions from "who governs" to "how they govern", distinguishing the agent from the architecture of their authority.

To bridge this conceptual gap, we define founder control as a governance configuration characterized by the capacity of founders to exert decisive, persistent influence over a firm's foundational commitments. Drawing on the Schumpeterian insight that the entrepreneur's essential function is to disrupt economic equilibria through creative destruction (Schumpeter, 1947), we posit that founder control is the critical instrument to enact this function within the firm. When this control structure enables an entrepreneurial ethos to permeate firm strategy, it combines the visionary foresight to identify novel opportunities with the operational resolve to commit the firm to their uncertain pursuit. Crucially, this conceptualization necessitates a clear distinction from the mere fact of founder ownership. While ownership can provide the formal basis for authority (*de jure* power), founder control denotes its actual exercise in shaping critical resource allocation decisions (*de facto* power), aligning with the classic organizational distinction between formal and real authority (Aghion & Tirole, 1997). This view thus shifts the analytical lens from static attributes like ownership or identity to the dynamic exercise of architectural influence, providing a structural view for understanding how sustained founder involvement becomes a productive force.

This study investigates how founder control shapes breakthrough innovation. Our analysis draws on a large panel of Chinese A-share listed firms in high-tech sectors from 2008 to 2022. We define founder control as the equity stake held by founders who remain actively engaged in firm governance, a measure designed to capture their enduring commitment to steering long-term technological strategy. To measure breakthrough innovation, we move beyond conventional metrics and leverage the textual content of patents. Following Kelly et al. (2021), we use SBERT embeddings to quantify each patent's technological originality (via backward textual similarity) and its forward influence (via forward textual similarity).

Our empirical strategy employs a lead-lag specification within a Poisson pseudo-maximum likelihood (PPML) framework to address the count nature of patent data. The baseline results reveal that founder control is positively associated with future breakthrough innovations, a relationship that is both robust and economically significant. To substantiate a causal interpretation, we subject this finding to a multi-dimensional identification strategy. This includes using unexpected founder departures as an instrument to isolate plausibly exogenous variation in founder control, a Heckman model to correct for selection bias, propensity score matching to construct a more rigorous counterfactual, and a series of targeted tests to rule out reverse causality. The main findings remain consistent across this suite of tests and are robust to a wide array of alternative measures, model specifications, and subsample analysis.

Beyond establishing a baseline association, we explore how founder control fosters breakthrough innovation. We propose two internal mechanisms and two external mechanisms. Internally, founder control enhances the firm's intrinsic capacity for innovation. It facilitates more intensive talent investment, which enables the concentration of R&D resources on a selected group of elite talent. This empowerment of a core team is vital for overcoming complex innovation challenges that might otherwise be intractable with dispersed resources. In parallel, founder control can impose rigorous internal discipline, which curtails the misallocation of valuable resources to unproductive avenues, thereby sharpening the focus on the innovative pursuit. Externally, founder control determines the firm's responsiveness to environmental shifts. It enables the firm to proactively reorient its strategy in response to industry turbulence. This allows the firm to identify and seize the niche markets within the narrow window of opportunity, turning high uncertainty into a distinct innovative edge. Furthermore, this responsiveness can be manifested as technological resilience against geopolitical blockades. Such resilience equips the firm not only to withstand external sustained pressures but also to catalyze indigenous innovation, demonstrating a fortitude forged through adversity. Collectively, these findings present an integrated framework of the founder-led innovation, detailing the distinct mechanisms through which founder control fosters breakthrough innovation.

To delineate the boundary conditions of our main effect, we examine several sources of heterogeneity. We first find that the innovation effect of founder control is more pronounced in firms led by single founders or small founding teams (≤ 4 members). Turning to internal architecture, we find the effect is contingent on the team's composition. For single-founder firms, where the challenge is to align vision with execution, the effect is more pronounced for founders who both consolidate formal authority (as Chairman-CEO) and possess broad functional expertise. Conversely, for multi-founder teams, where the challenge is to mitigate conflict and enable action, the effect is more pronounced under a clear internal hierarchy, namely concentrated equity ownership and functional specialization. Beyond these structural conditions, the innovation effect is amplified through complementarity with specific founder attributes, including demographic profile, formative experiences, and positional role within the firm. Finally, we analyze the temporal dynamics of this relationship, documenting an inverted U-shaped pattern between founder tenure and breakthrough innovation. For founder-CEOs with actual control, the peak of this innovation lifecycle occurs substantially later, reinforcing the importance of sustained authority.

Our paper makes several contributions. The first is to extend the literature on founder governance and corporate innovation, where the debate has largely been confined to the relative merits of founder-led versus professionally managed firms (Chahine & Zhang, 2020; Kannan-Narasimhan et al., 2023) or a focus on static founder attributes (Dawson et al., 2018; Pan et al., 2024). We move this conversation forward by proposing and empirically validating the construct of founder control, which is defined as the dynamic capacity to shape a firm's innovation strategy rather than by founder identity or ownership alone. This shift in perspective reframes the central question from who governs to how specific governance structures shape a firm's capacity for proactive exploration under

uncertainty. We substantiate this perspective by investigating the underlying mechanisms through which founder control operates, examining its effects on both internal governance and external adaptation. Finally, we explore key heterogeneous effects, demonstrating that the innovation-enhancing effects of founder control are contingent upon the structure and attributes of the founding team. Together, these findings provide a more detailed understanding of when and why this governance model fosters breakthrough innovation.

Second, we make a methodological contribution by developing a text-based measure of breakthrough innovation using advanced natural language processing. While prior studies have established a valuable foundation using metrics such as citation counts and patent types (Acemoglu et al., 2022; Kraft et al., 2025), our approach aligns with a nascent but influential stream of research that leverages patent text to capture an innovation's originality and impact. Specifically, we build upon and extend the conceptual approach of Kelly et al. (2021) by operationalizing breakthrough innovation along two theoretically salient dimensions: technological originality (measured as semantic dissimilarity to prior patents) and pathway influence (measured as semantic similarity to subsequent patents). Using SBERT-based text embeddings, our approach captures the novelty and impact of each patent, allowing for a more precise, scalable, and sophisticated measurement of breakthrough innovation. This methodology advances the empirical toolkit available for studying the antecedents of radical and transformative innovation.

Finally, we offer timely insight into the potential outcomes of the landmark institutional opening-up that formally introduced control-enhancing mechanisms nationwide. While prior research from developed economies often questioned concentrated founder control (Aggarwal et al., 2022; Field & Lowry, 2022), we argue that it acts as a powerful internal governance substitute in the context of weaker external governance and higher contracting uncertainty. This internal governance provides the stability and long-term orientation necessary to shield high-risk innovation from market pressure. Our findings therefore suggest that for emerging markets, empowering founders is a potent strategic solution to overcome institutional voids and foster breakthrough innovation.

2. Literature and hypothesis

2.1. Literature review

2.1.1. The governance challenge of breakthrough innovation

Breakthrough innovations, defined as novel technologies or products that substantively alter a market's competitive landscape, are a primary engine of long-run economic growth (Schumpeter, 1934). Such innovations are distinct from incremental innovation that refines existing capabilities. They are characterized by high technical and commercial uncertainty, long and variable investment horizons, and a significant probability of complete failure (Byun et al., 2021). Consequently, the central challenge is not simply funding R&D, but creating a governance environment that can successfully manage the difficult process of exploration.

The first governance challenge is a temporal mismatch of incentives. Breakthrough projects require sustained and substantial capital expenditure without yielding immediate returns, yet managerial careers and compensation are typically tied to short-term verifiable performance metrics (Manso, 2011). The pressure is particularly acute in public firms, where market scrutiny can systematically curtail investment in novel and long-gestation projects (Bernstein, 2015). This environment reinforces a natural organizational tendency to favor exploitation over exploration. Rational managers will systematically prioritize predictable incremental projects over ambitious exploratory ventures. Overcoming this deep-seated institutional bias requires more than patience; it requires a governance structure that fosters a genuine "tolerance for failure" (Tian & Wang, 2014). The ultimate challenge is to create an internal architecture that actively rewards the intelligent pursuit of ambitious uncertainty, shielding exploratory projects from premature termination and providing the patient capital necessary to navigate the uncertain path to discovery.

The second relates to the nature of decision-making authority. In professionally managed firms, formalized capital budgeting systems like discounted cash flow analysis are a cornerstone of governance. They are designed to impose procedural discipline, mitigate agency costs, and ensure capital flows to projects with the highest verifiable returns (Chen et al., 2011). While rational for managing existing assets, this logic is often epistemologically misaligned with breakthrough innovation. The potential of such projects cannot be captured by quantifiable near-term forecasts. Instead, it relies on qualitative market judgment and foresight into a future state that does not yet exist (Das et al., 2017). Consequently, the very protocols designed to ensure efficiency become structural barriers that systematically filter out the most radical yet potentially most valuable ideas. The challenge therefore transcends finding a project champion. It requires a governance structure that grants key actors the legitimate authority to make belief-based bets, effectively overriding the firm's established analytical machinery.

These challenges highlight that a firm's capacity for breakthrough innovation is contingent on its ability to solve a specific set of governance problems. It must find a way to insulate its innovation strategy from myopic pressures, tolerate failure as a necessary part of exploration, and empower a decision-making authority with both a long investment horizon and a unique objective function. This raises the critical question of what specific governance structure can fulfill these demanding requirements.

2.1.2. The competing incentives of founder control

Within the spectrum of governance structures, founder control offers a particularly sharp illustration of this governance challenge. The structure creates powerful but opposing incentives regarding a firm's commitment to breakthrough innovation, leading to a deep theoretical tension.

One perspective, consistent with the logic of stewardship theory (Davis et al., 1997), posits that founders are particularly capable of overcoming the contracting problems. Unlike professional agents, founders often act as stewards whose utility is inextricably linked to the firm's perpetuity and trans-generational ambition, a concept captured by the notion of socioemotional wealth (Gómez-Mejía et al.,

2007). This generational orientation provides a natural antidote to managerial myopia. Moreover, the concentration of decision-making authority allows founders to insulate the innovation process from conventional resource-allocation discipline. They can underwrite high-uncertainty projects based on a coherent entrepreneurial spirit, thereby approving the very exploratory paths that formalized analytical procedures are designed to reject.

An opposing line of reasoning, grounded in upper echelons theory (Hambrick, 2007), posits that organizational outcomes are reflections of the values, cognitive bases, and characteristics of top executives. Founders are not a homogeneous group. They exhibit significant heterogeneity in their personal attributes and managerial philosophies. While some founders are transformative leaders who aggressively pursue innovative endeavors, others may be more conservative (Ardito et al., 2025). Especially in later-stage or multi-generational family firms, the founder's dominant motivation may shift from entrepreneurial value creation to the preservation of family wealth and status, increasing resistance to disruptive innovation (Blanco-Mazagatos et al., 2022). Additionally, in industries with stable technological paradigms, the particular temporal focus of founders whose identities are closely tied to the firm's established products or processes may resist new product innovation that could undermine their legacy (Nadkarni & Chen, 2014). This entrenchment can stifle innovation, as the founder's influence may suppress emerging ideas that challenge the status quo.

These competing arguments create a significant theoretical ambiguity. The incentives generated by founder control can either align the firm with the pursuit of high-risk projects or entrench a risk-averse strategy, rendering the net effect theoretically indeterminate. This paper aims to address this ambiguity by proposing and testing the conditions that tilt this balance.

2.2. Hypothesis development

While the net effect of founder control is theoretically ambiguous, a compelling stream of empirical evidence offers a clear direction. Studies consistently find that founder-led firms systematically outperform their manager-led counterparts in generating productive value and pursuing innovation-focused strategies, an advantage particularly pronounced in highly innovative industries (Harrison et al., 2024; Lee et al., 2020; Pan et al., 2024). This consistent empirical pattern thus directs the theoretical inquiry to explain *why* the innovation-driving advantages of founder control consistently prevail over its liabilities in specific contexts like the high-tech sector. Our central argument posits that this capacity for breakthrough innovation is determined not by the leader's identity per se (founder vs. manager), but by the governance architecture that empowers an entrepreneurial founder. Here, the term "entrepreneurial" signifies a distinct orientation aimed at transformative value creation, a quality separate from the mere fact of company creation or the stewardship of existing assets (Seyb et al., 2025). We propose that the control exercised by such founders fosters breakthrough innovation because it allows two of their fundamental characteristics to shape firm strategy: a long-term vision under deep uncertainty and an enduring commitment to organizational coherence.

First, such founders are distinguished by the formulation of a strategic vision. This concept transcends the passive recognition of opportunities; it describes the articulation of an ambitious competitive agenda that stretches a firm far beyond its current resource endowment (Preller et al., 2020). Their intent is born from a non-consensus conception of the future and a profound ambition to reshape market boundaries. It provides the directional clarity and motivational force necessary to mobilize an organization towards a goal that would be systematically filtered out by more conventional governance structures that prioritize near-term predictability (Abebe & Tangpong, 2018). Therefore, such vision initiates the Schumpeterian process of creative destruction from a place of deliberate purpose rather than serendipitous discovery.

Second, this strategic vision needs to be operationalized through the imposition of a disciplined resource architecture. Entrepreneurial founders act as a vanguard of a specific, coherent, and often austere model of resource configuration designed to realize their vision (Fauchart & Gruber, 2011). This involves a rigorous prioritization of indispensable inputs such as brilliant talent and specialized capital for the transformative goal and a corresponding intolerance for innovation failure (Chang et al., 2024). This governance model ensures that the firm's operational reality is tightly aligned with its intent, creating an environment where resources are not just accumulated but are productively focused (Bernstein et al., 2017).

These defining characteristics, however, remain inert without the structural power to enact them. Founder control is the critical instrument that translates the founder's personal vision and organizational philosophy into corporate practices, thereby offering a direct solution to the two core governance challenges of breakthrough innovation. The vision provides the crucial insulation from the external pressures for conformity and the internal tendencies towards resource inertia that typically afflict firms with dispersed ownership (Asker et al., 2015). The autonomy empowers the founder to make the patient, high-stakes, and often irreversible investments that are the hallmark of breakthrough innovation, ensuring that the firm's technological trajectory remains aligned with their long-term intent, even in the face of uncertainty and short-term performance volatility. This leads to our baseline hypothesis:

Hypothesis 1. Founder control is positively associated with breakthrough innovation.

2.3. Mechanism analysis

To further elucidate how founder control translates into breakthrough innovation, we propose four theoretically plausible mechanisms: two internal to the firm and two related to its external posture.

The first mechanism is a strategy of intensive talent investment. This approach rejects the conventional managerial logic of risk mitigation through resource diffusion, where capital is spread thinly across numerous projects and teams (Gormley & Matsa, 2016). Instead, the entrepreneurial founder with technical passion will exhibit a clear tendency to concentrate resources on a few high-impact bets. This tendency primarily manifests in two ways. The first aspect is attracting and retaining a small cadre of top-tier innovators.

Founders often function as a talent magnet to attract and retain exceptional human capital far beyond the capacity of formal incentive systems (Kaehr Serra & Thiel, 2019). The second, and more critical, aspect is allocating R&D funds directly to this core talent base. Such allocations amplify the productivity of each individual innovator by creating an environment of extreme resource density. In this way, the founder assembles a focused elite team to address challenges whose complexity and scale that are inaccessible to larger but more diffusely resourced competitors (Jaravel et al., 2018). This allocative concentration of R&D resources is a critical act that dramatically raises the probability of achieving a genuine technological breakthrough. Thus, we hypothesize:

Hypothesis 2a. Founder control fosters breakthrough innovation by facilitating intensive talent investment.

The second mechanism is the imposition of a rigorous internal discipline. This discipline counters two distinct forms of organizational entropy that inhibit innovation. Manager-led firms are often susceptible to bureaucratic bloat and managerial entrenchment (Bates et al., 2008), while less ambitious founder-led firms can succumb to a complacent inertia that allows organizational slack to accumulate (Block, 2012). Guided by a singular, ambitious vision, the disciplined founders actively combat both tendencies. Their primary instrument is a mission-oriented culture that aligns the organization and provides the legitimacy for a governance model of operational parsimony. This parsimony involves maintaining lean organizational structures, minimizing administrative layers, and demonstrating a low tolerance for any expenditure that does not directly contribute to the core innovation objective (Marquis & Tilcsik, 2013). This austerity is not an end in itself but a managerial necessity, reflecting a sovereign prerogative where any squandered resource is viewed as a direct impediment to achieving the long-term goal (Wasserman, 2017). By enforcing this rigorous internal discipline, the founder ensures that the firm's resources are efficiently used and channeled towards the innovative activities that create future value, rather than being dissipated by administrative friction or managerial indulgence. Thus, we hypothesize:

Hypothesis 2b. Founder control fosters breakthrough innovation by enabling the imposition of rigorous internal discipline.

The third mechanism concerns the firm's strategic reorientation, namely the capacity to execute a fundamental shift in the firm's trajectory in response to external uncertainty. In innovation-intensive sectors, such uncertainty often manifests as industry turbulence, a state of flux characterized by dramatically changing market leaders and high rates of firm entry and exit. Breakthrough innovation often originates from interpreting and acting upon ambiguous signals of an emergent technological paradigm (Kivimaa & Kern, 2016). Manager-led firms or even conservative founder-led firms typically respond with caution. While prudent in a stable environment, this deliberate pace may become a disadvantage when windows of opportunity are fleeting (Maine et al., 2015). Control of an entrepreneurial founder, in contrast, enables a different logic. It allows the firm's technology choices to be anchored not in consensus or objective proof, but in the founder's deep conviction and unique understanding of the nascent opportunity. This autonomy enables the founder to execute a decisive reorientation of the firm's resources (Teece, 2007), committing the firm to a new trajectory long before its logic is widely recognized. It is this proactivity of strategic commitment amidst uncertainty, while competitors wait for clear signals, that positions the firm to capitalize on paradigm shifts and achieve breakthrough innovation. Thus, we hypothesize:

Hypothesis 3a. Founder control fosters breakthrough innovation through strategic reorientation in response to industry turbulence.

The fourth and final mechanism is technological resilience. Breakthrough innovation often provokes strong opposition from incumbents or geopolitical rivals seeking to protect their economic rents (Akcigit & Van Reenen, 2023). Such opposition may include disrupting critical supply chains or denying access to essential technologies, pressures designed to halt a firm's progress. A manager-led firm, bound by its fiduciary duty to de-risk (He & Tian, 2013), will often retreat from the contested technological domains or even abandon them entirely. Founder control, however, provides the insulation to withstand these pressures. For ambitious founders, whose

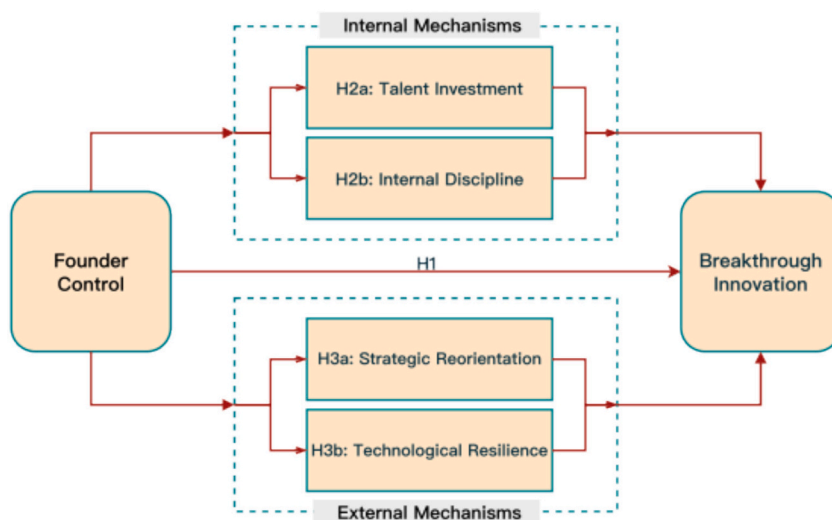


Fig. 1. Theoretical framework.

career mission is frequently inseparable from their personal identity, external challenges are not insurmountable risks but tests of resolve. This personal commitment enables them to maintain a steadfast course, absorb near-term losses, and intensify efforts to overcome external dependencies. This fortitude enables the firm to persevere on a contested path long after a risk-averse competitor would have capitulated (Faleye et al., 2014). Thus, we hypothesize:

Hypothesis 3b. Founder control fosters breakthrough innovation through technological resilience against external opposition.

To synthesize the above arguments, the theoretical framework is presented in Fig. 1.

3. Methodology

3.1. Data and sample

Our empirical analysis relies on a panel dataset of publicly listed high-tech firms on China's A-share market, spanning from 2008 to 2022. We choose this timeframe to establish a stable institutional baseline essential for our identification strategy. The start year, 2008, post-dates two critical market-wide reforms. First, by this point, the vast majority of Chinese firms had completed the split-share structure reform (Tan et al., 2020), a fundamental shift that largely aligned the economic interests of tradable and non-tradable shareholders, thereby stabilizing a core mechanism of corporate governance. Second, the enhanced disclosure standards mandated by the China Securities Regulatory Commission in 2007 (e.g., Rule No. 15) had taken full effect, significantly improving the quality and comparability of firm-level data, particularly concerning innovation-related expenditures. Starting our analysis in 2008 thus mitigates significant confounding effects from these foundational institutional shifts. The sample period concludes in 2022 to ensure that our patent-based measurement of breakthrough innovation, which often involves multi-year administrative lags, is not systematically truncated. This cutoff point minimizes right-censoring issues and avoids the underestimation of a firm's innovation output.

Our theoretical focus on breakthrough innovation guides the construction of our sample from technology- and knowledge-intensive industries. We define these industries by mapping the core industrial groupings from three official National Bureau of Statistics (NBS) classifications to the industry codes used for listed firms. The source documents include *The Classification of High-Tech Industries (Manufacturing Sector) (2017)*, *The Classification of High-Tech Industries (Service Sector) (2018)*, and *The Classification of Intellectual Property (Patent)-Intensive Industries (2019)*. They specify industries at the 4-digit level, while the standard classification for listed firms is at the 2-digit level of the same national standard (GB/T 4754–2017). Our process therefore captures the 2-digit industries where the NBS-designated 4-digit sub-industries are demonstrably concentrated, while setting aside those where the connection to the official classifications rests on isolated or anomalous sub-categories not representative of the industry's primary activities. The resulting sample is thus concentrated in 15 core industries (C26–C27, C34–C40, I63–I65, and M73–M75), encompassing sectors such as pharmaceuticals, computer and telecommunications equipment, and software and information technology services.

Building on this sample of firms in high-tech industries, we apply several additional firm-level filters. First, we restrict the sample to firms with identifiable founders. This step excludes firms whose ownership originates from state-controlled entities or the restructuring of collectively-owned enterprises, which by nature lack a founding entrepreneur and thus fall outside our theoretical scope. Second, we remove all eight firms listed on the STAR Market with DCS structure. The limited number and their brief post-listing window (primarily post-2020) cannot yield direct evidence nor robust inference under our 3-year lead-lag specification. Third, we remove firm-year observations for firms designated as "Special Treatment" (ST) or "Particular Transfer" (PT), as their distressed financial status introduces confounding operational behaviors. We also exclude firms that underwent a major asset restructuring or changed their primary industry classification to ensure the consistency of our panel data. Fourth, we exclude observations with incomplete financial data for the variables required in our empirical models. Our final sample comprises 6092 firm-year observations from 930 unique firms.

The data for this study are compiled from three primary sources. Firm-level financial data are obtained from the Wind database. Data on corporate governance, including founder identification and ownership structures, are sourced from the China Stock Market & Accounting Research (CSMAR) database and manually cross-validated against firms' annual reports for accuracy. Patent data are obtained from the official database of the China National Intellectual Property Administration (CNIPA).

3.2. Variables

3.2.1. Independent variable: founder control

A persistent challenge in governance research is operationalizing control to capture not only formal authority but also the idiosyncratic, long-horizon objectives of key shareholders. This is particularly acute in firms under founder stewardship, where fundamental imperatives are often shaped by the pursuit of non-financial goals such as legacy and autonomy, collectively termed socioemotional wealth (Gómez-Mejía et al., 2007). While common governance proxies such as founder-CEO duality, founder tenure, or board representation offer valuable insights (Gao & Jain, 2012), they often provide an incomplete picture. These measures can be binary or indirect, failing to fully capture the gradations of influence rooted in a founder's unique history and, most critically, their personal capital commitment. This highlights the need for a measure that faithfully represents a founder's structural capacity to shield these long-horizon ambitions from external pressures.

Given the limitations of agent proxies, we posit that this capacity is most directly and powerfully captured by the founder's direct equity ownership. In contrast to complex control structures (e.g., pyramids/DCS), direct ownership represents a clear and substantial personal commitment that functions on two levels. First, it creates an unambiguous alignment of the founder's personal financial

reward with the firm's innovative trajectory. Second, and more importantly, this “skin in the game” serves as a costly and credible signal of the founder's internal belief in their technological choice. It is this high-stakes alignment that confers the informal authority and structural leverage necessary to pursue ambitious innovation agendas and to withstand pressure for premature concessions.

Following this logic, we operationalize founder control as the aggregate equity stake of all founders actively engaged as directors or executives in a given firm-year. While China's corporate governance framework defines several discrete control thresholds such as 67 % (supermajority), 51 % (simple majority), and 34 % (veto power), we argue that a founder's ability to shape strategy is not merely a step function of these legal rights. Instead, their influence, arising from enhanced bargaining power and moral authority, grows continuously with their financial stake. A continuous variable is therefore better suited for our baseline model, as it captures the full variation in the data without imposing a restrictive functional form. To ground our analysis in these institutional realities, we test the importance of these critical thresholds in our robustness checks. This approach assumes coherence within the founding team (Villalonga & Amit, 2006). We explore the internal architecture of the founding team as a source of heterogeneity in Section 7.

3.2.2. Dependent variable: breakthrough innovation

Existing measures of breakthrough innovation, often based on citation counts (Kraft et al., 2025), can be confounded by temporal and inter-industry biases. We therefore develop a measure that directly captures an innovation's two defining characteristics: its technological originality and its forward path-defining influence. Accordingly, we conceptualize a breakthrough innovation as one that simultaneously represents a significant departure from prior art (originality) and serves as a critical foundation for subsequent technological developments (influence).

Our measurement process begins with the construction of a targeted patent pool. We identify patents granted to Chinese A-share listed firms within significant domains by matching International Patent Classification (IPC) codes against authoritative frameworks from China's Ministry of Science and Technology (MOST), specifically the *Key Digital Technology Patent Classification System (2023)* and the *Classification Correspondence Table between Strategic Emerging Industries and International Patent Classification (IPC) (2021)*. To ensure our sample consists of patents with at least a baseline of technological relevance, we then refine this pool by excluding patents that remain unreferenced throughout our observation period.

To operationalize originality and influence, we follow and extend the textual analysis approach of Kelly et al. (2021). Using a fine-tuned Sentence-BERT (SBERT) model, we transform the textual content (title, abstract, claims) of each patent into a high-dimensional vector embedding v_k . For each focal patent, we calculate its backward similarity S_b as the mean cosine similarity to patents in its 4-digit IPC field from the preceding five years, and its forward similarity S_f as the mean cosine similarity to patents in its field from the subsequent five years. These scores are defined as follows:

$$S_b = \frac{1}{N_b} \sum_{i=1}^{N_b} \cos(v_k v_i^b), S_f = \frac{1}{N_f} \sum_{j=1}^{N_f} \cos(v_k v_j^f), \quad (1)$$

where v_k is the focal patent's embedding vector; v_i^b and v_j^f are the embedding vectors of the i -th backward patent and the j -th forward patent, respectively; and N_b and N_f are the counts of those patents.

Since innovation dynamics vary across fields, we normalize these raw scores by the average similarity within their respective 4-digit IPC fields. Let P denote the number of patents in the focal patent's IPC field. The industry-adjusted scores, $\bar{S}_b$ and $\bar{S}_f$ are defined as:

$$\bar{S}_b = \frac{S_b}{\frac{1}{P} \sum_{p=1}^P S_{b,p}}, \bar{S}_f = \frac{S_f}{\frac{1}{P} \sum_{p=1}^P S_{f,p}}, \quad (2)$$

where $p \in 1, \dots, P$ denotes the p -th patent within the field as the focal patent. These adjusted scores represent a patent's relative originality and influence.

We classify a patent as a breakthrough if it meets stringent criteria for both originality and influence, based on its position in the field-specific empirical cumulative distribution functions of these adjusted scores. The cumulative distribution functions for adjusted backward and forward similarity scores, $F_b(\bar{S}_b)$ and $F_f(\bar{S}_f)$, are formally defined as:

$$F_b(s) = \frac{1}{P} \sum_{p=1}^P I(\bar{S}_{b,p} \leq s), F_f(s) = \frac{1}{P} \sum_{p=1}^P I(\bar{S}_{f,p} \leq s). \quad (3)$$

A patent is classified as a breakthrough if it is an outlier on both dimensions. We operationalize this by setting stringent quantile thresholds: a patent's backward similarity must fall below the 20th percentile ($Q_{b,20}$) and its forward similarity must exceed the 80th percentile ($Q_{f,80}$). This dual-quintile requirement ensures that we identify only those innovations that are both radically novel relative to their technological antecedents and disproportionately influential on subsequent developments. The quantile thresholds are formally expressed as:

$$Q_{b,20} = \inf\{s | F_b(s) \geq 0.2\}, Q_{f,80} = \inf\{s | F_f(s) \geq 0.8\}. \quad (4)$$

The identification rule is expressed as:

$$I(\text{Breakthrough}) = I(\bar{S}_b < Q_{b,20}, \bar{S}_f > Q_{f,80}). \quad (5)$$

Finally, we aggregate these breakthrough patents at the firm-year level to construct our dependent variable. In particular, we include only those patent applications that are eventually granted. This requirement serves a dual purpose. First, it ensures that our measure reflects externally validated innovation rather than mere inventive attempts. Second, it addresses the concern that firms may strategically file patents for window-dressing purposes, which would otherwise inflate measures of innovative outputs. Consequently, this carefully specified variable captures a firm's output of truly substantive and impactful innovation.

3.3. Empirical strategy

The statistical properties of patent data, which are non-negative integer counts with a preponderance of zeros, render standard linear models unsuitable. Ordinary Least Squares (OLS) is ill-suited because it can generate illogical out-of-range predictions and fails to account for the heteroskedasticity inherent in count data. The common alternative of a log-transformation is also untenable, as it systematically biases the estimates of the original conditional mean (Silva & Tenreiro, 2006). We therefore employ a PPML estimator. This method is particularly appropriate for our context because it correctly handles the non-negative count data and, crucially, provides consistent estimates under the relatively weak assumption that the conditional mean is correctly specified (Chen & Roth, 2024). We specify the conditional mean as follows:

$$\mathbb{E}(\text{Patents}_{it} | FC_{it}, \mathbf{X}_{it}, \eta_i, \delta_t) = \exp(\beta_1 FC_{it} + \mathbf{X}_{it}' \gamma + \eta_i + \delta_t), \quad (6)$$

where Patents_{it} represents the number of breakthrough patent applications filed by firm i in year t . Acknowledging the significant gestation period required for transformative R&D in high-technology industries (Tan et al., 2020), we introduce a temporal lag between our independent variables (year t) and the innovation outcome (year τ), setting $\tau = t + k$. Following the established practice by Hall et al. (2001), we adopt a three-year “safety lag” ($k = 3$) for our baseline specification. This lead-lag structure is a critical design feature to mitigate concerns of reverse causality.

The key independent variable, FC_{it} , represents founder control at firm i in year t . $\mathbf{X}_{it}$ includes a comprehensive set of controls for observable firm characteristics that could otherwise confound the estimated relationship, following established practice in the innovation literature (Acemoglu et al., 2022; Tian et al., 2024). Our identification strategy relies on firm (η_i) and year (δ_t) fixed effects to control for all time-invariant unobserved firm heterogeneity and macro-level shocks, respectively. All continuous control variables are winsorized at the 1 % and 99 % levels to mitigate the influence of extreme outliers. Standard errors are clustered at the firm level to account for potential serial correlation. Detailed definitions for all variables are provided in Table 1.

3.4. Summary statistics

Table 2 presents the summary statistics for the main variables. Panel A reports the distributional properties for the full sample of 6092 firm-year observations. Our dependent variable, Patents_{t+3} , reflects the stochasticity and rarity of high-impact innovation. The variable's mean is 1.411, while both the median and the 75th percentile are 1.000, resulting in a markedly right-skewed distribution with a standard deviation (2.861) more than double the mean. This pronounced skewness and the count nature of the data provide a strong statistical rationale for our choice of a PPML model.

Our key independent variable, founder control (FC), has a sample mean of 0.239 and a standard deviation of 0.186, indicating substantial cross-sectional heterogeneity. The interquartile range from 0.054 to 0.380 further shows a wide spectrum of founder control, which is essential for identifying the effects of founder-centric governance. The control variables suggest the sample consists of established, profitable, and moderately leveraged high-tech firms. Overall, these statistics confirm sufficient variation in both governance structures and innovation outcomes for the analysis.

Importantly, Panel B decomposes the variance of FC to directly address its temporal dynamics. It reveals that while a large portion

Table 1
Variable definitions.

Label	Name	Definition
<i>Patents</i>	Breakthrough innovation	The number of patent applications filed in year t that are eventually granted and are classified as breakthrough based on backward and forward similarities.
<i>FC</i>	Founder control	The cumulative proportion of equity ownership held by individuals identified as firm founders.
<i>R&D</i>	R&D expenses	The natural logarithm of R&D expenses.
<i>Size</i>	Firm size	The natural logarithm of total assets.
<i>Age</i>	Firm age	The natural logarithm of years since the firm's IPO.
<i>Roa</i>	Return on assets	Net income divided by total assets.
<i>Current</i>	Current ratio	Current assets divided by current liabilities.
<i>Lev</i>	Leverage	Total debt divided by total assets.
<i>Bm</i>	Book-to-market ratio	Book value of equity divided by market value of equity.
<i>Top1</i>	Largest shareholder ownership	The percentage of shares owned by the largest shareholder.
<i>Balance</i>	Ownership balance	The ratio of the largest shareholder's ownership to the combined ownership of the 2nd to 10th largest shareholders.
<i>Board</i>	Board size	The natural logarithm of the total number of directors on the board.
<i>Inst</i>	Institutional ownership	The percentage of shares owned by institutional investors.
<i>Tmtage</i>	TMT age	The average age of top management team members.

Table 2
Summary statistics.

Panel A: Full sample										
	N	Mean	S.D.	Min	P10	P25	Median	P75	P90	Max
<i>Patents_{t+3}</i>	6092	1.411	2.861	0.000	0.000	0.000	1.000	1.000	3.000	18.000
<i>FC</i>	6092	0.239	0.186	0.000	0.000	0.054	0.228	0.380	0.504	0.632
<i>R&D</i>	6092	17.165	3.323	0.000	16.328	16.950	17.649	18.347	19.050	20.628
<i>Size</i>	6092	21.418	0.869	19.695	20.377	20.758	21.331	21.964	22.583	23.890
<i>Age</i>	6092	1.332	0.747	0.000	0.000	0.693	1.386	1.946	2.197	2.708
<i>Roa</i>	6092	0.052	0.058	−0.310	0.009	0.028	0.053	0.080	0.110	0.184
<i>Current</i>	6092	3.809	3.568	0.648	1.192	1.589	2.442	4.482	8.256	17.004
<i>Lev</i>	6092	0.309	0.167	0.042	0.100	0.172	0.292	0.428	0.546	0.721
<i>Bm</i>	6092	0.370	0.154	0.066	0.179	0.255	0.354	0.475	0.591	0.725
<i>Top1</i>	6092	32.683	13.269	8.927	16.974	22.308	30.740	41.277	51.619	63.750
<i>Balance</i>	6092	0.895	0.625	0.075	0.218	0.405	0.738	1.243	1.838	2.624
<i>Board</i>	6092	2.204	0.160	1.792	1.946	2.079	2.303	2.303	2.303	2.485
<i>Inst</i>	6092	31.054	24.940	0.112	3.170	9.453	23.291	50.952	70.050	84.510
<i>Tmtime</i>	6092	47.783	3.125	40.570	43.650	45.640	47.750	49.910	51.820	55.060

Panel B: Decomposition of variance for founder control						
Variable		Mean	S.D.	Min	Max	Observations
<i>FC</i>	Overall	0.239	0.186	0	0.632	<i>N</i> = 6092
	Between		0.182	0	0.632	<i>n</i> = 930
	Within		0.060	−0.272	0.660	<i>T</i> = 6.551

of the total variation is between firms (Between s.d. = 0.182), there is also meaningful variation within firms over time (Within s.d. = 0.060). This finding demonstrates that founder control is not a static firm attribute but rather a dynamic variable that evolves, for instance, through founders' equity transactions or changes in their active management roles. This very within-firm variation is the source of identification for our primary empirical strategy, which employs firm fixed-effects models to purge time-invariant

Table 3
Baseline regression results.

Dependent variable: <i>Patents_{t+3}</i>				
	(1)	(2)	(3)	(4)
<i>FC</i>	0.901*** (0.160)	0.853*** (0.188)	0.775*** (0.135)	0.761*** (0.137)
<i>R&D</i>	0.044*** (0.010)	0.063*** (0.003)	0.001 (0.005)	0.007*** (0.002)
<i>Size</i>	0.304*** (0.048)	0.434*** (0.062)	0.062* (0.037)	0.168*** (0.029)
<i>Age</i>	−0.080* (0.043)	−0.154*** (0.057)	0.083 (0.053)	0.076 (0.115)
<i>Roa</i>	3.036*** (0.753)	1.733*** (0.595)	0.824*** (0.162)	0.620*** (0.205)
<i>Current</i>	0.048*** (0.013)	0.028** (0.013)	0.032*** (0.004)	0.020*** (0.005)
<i>Lev</i>	0.743*** (0.173)	−0.201 (0.175)	0.103 (0.079)	−0.262** (0.118)
<i>Bm</i>	−0.735*** (0.133)	−1.849*** (0.230)	0.371*** (0.136)	−0.029 (0.285)
<i>Top1</i>	−0.007 (0.006)	−0.006 (0.006)	0.018*** (0.003)	0.011** (0.006)
<i>Balance</i>	−0.118 (0.163)	−0.128 (0.156)	0.129* (0.067)	0.057 (0.053)
<i>Board</i>	0.234 (0.247)	−0.000 (0.263)	0.500*** (0.184)	0.332 (0.247)
<i>Inst</i>	0.007*** (0.002)	0.004 (0.003)	0.006* (0.004)	0.003 (0.004)
<i>Tmtime</i>	−0.056*** (0.011)	−0.035*** (0.011)	0.013*** (0.004)	0.022*** (0.006)
Firm FE	Yes	No	Yes	Yes
Year FE	Yes	Yes	No	Yes
Pseudo <i>R</i> ²	0.066	0.093	0.434	0.444
<i>N</i>	6092	6092	6092	6092

Note: Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

unobserved heterogeneity.

4. Results

4.1. Baseline results

Table 3 presents the baseline estimation results examining the relationship between founder control and future breakthrough innovation. Column (1) establishes the baseline cross-sectional correlation with a full set of controls. Column (2) then introduces year fixed effects to account for aggregate temporal shocks, while Column (3) instead uses firm fixed effects to control for time-invariant unobserved heterogeneity. The final specification in Column (4) incorporates both sets of fixed effects, thereby isolating the association from these confounding factors simultaneously. Across all specifications, the coefficient on founder control remains positive and highly significant.

The economic magnitude implied by this association is also substantive. The coefficient of 0.761 in the full specification in Column (4) indicates that a one-standard-deviation increase in founder control is associated with a 0.142 increase in the count of breakthrough patents three years later, representing a 10.03 % increase relative to the sample mean. Thus, Hypothesis 1 is supported.

While these results establish a robust conditional correlation, they do not substantiate a causal claim. We therefore proceed to address the potential endogeneity concerns in the subsequent analysis.

4.2. Endogeneity concerns

While our baseline model mitigates some endogeneity concerns by including firm and year fixed effects and a rich set of controls in a lead-lag specification, the causal interpretation of our findings warrants further scrutiny. Three primary threats to identification persist. First, omitted variable bias may arise from unobservable, time-invariant founder attributes, such as innate entrepreneurial talent or risk tolerance, which correlate with both the founder's equity control and the firm's pursuit of innovation. Second, selection bias is a concern if the firms that maintain founder control are systematically different in unobserved ways that also drive innovation outcomes. Third, the possibility of reverse causality exists, whereby successful innovation could in turn alter a founder's ownership stake, for instance, by increasing their bargaining power or personal wealth to buy back shares.

To substantiate our main findings, we systematically address these issues using a multi-pronged empirical strategy: (1) an instrumental variable approach to account for omitted variables, (2) a Heckman two-stage model to correct for potential selection bias, (3) propensity score matching to construct a more comparable control group, and (4) tests for reverse causality.

4.2.1. Instrumental variable approach

The estimated association between founder control and breakthrough innovation may suffer from endogeneity due to omitted variables that simultaneously influence both founder equity stakes and firms' innovative output. For instance, a firm on a promising technological trajectory might be more inclined to retain its founder's leadership, while a series of innovation failures could precipitate a founder's removal. To address these concerns and move towards a causal interpretation, we employ a two-stage least squares (2SLS) instrumental variable (IV) approach.

To instrument for founder control, we exploit variation stemming from the unexpected departure of a founder-executive. We define

Table 4

Endogeneity tests I: instrumental variable approach and heckman model.

	IV-2SLS		Heckman model
	First stage (1)	Second stage (2)	Outcome equation (3)
<i>Depart</i>	−0.019** (0.005)		
<i>FC</i>		6.262** (2.787)	1.735*** (0.442)
<i>IMR</i>			0.928* (0.374)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Under-identification test	22.56***	-	-
Weak instrument test	24.91	-	-
χ^2	-	-	11,891.786***
N	6092	6092	4407

Note: This table reports results from two endogeneity tests. Columns (1) and (2) present the first- and second-stage results of a 2SLS instrumental variable estimation, respectively. The dependent variable is *FC* in Column (1) and *Patents*_{t+3} in Column (2). Column (3) reports the second-stage results from a Heckman selection model, where the dependent variable is *Patents*_{t+3}. *IMR* denotes the Inverse Mills Ratio calculated from the first-stage selection model. Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

such a departure as one arising from events that are plausibly exogenous to the firm's immediate performance or innovation strategy, including a founder's sudden death, debilitating health issues, or criminal charges unrelated to firm operations. When a founder-executive departs for such a reason in year t , our instrument, *Depart*, equals 1 for years t , $t + 1$, and $t + 2$, and 0 otherwise. This three-year window is designed to capture the prolonged transition period during which the governance structure and corporate direction of the firm adjust following the abrupt loss of a founder's influence, in line with Lee et al. (2020).

The validity of this instrument rests on two conditions. The relevance condition is met as the unexpected departure of a founder-executive directly weakens founder control. Since a founder-executive is a primary source of founder control, their sudden departures from a key leadership role precipitously strip them of the formal authority and informal influence. The exclusion restriction rests on the nature of the events. A sudden illness or fatal accident is inherently an external shock. Unlike a planned retirement or a performance-related dismissal, it is not an endogenous response to private information about the firm's internal R&D pipeline (e.g., the imminent success or failure of a key project). It therefore affects innovation outcomes only through its disruption of founder control.

Table 4 reports the 2SLS estimation results. The first-stage regression in Column (1) shows that our instrument has the expected negative and significant coefficient. The validity of this instrument is confirmed as the under-identification test statistic rejects the null hypothesis and the weak instrument test statistic surpasses conventional thresholds. The second-stage result in Column (2) reveals that instrumented founder control has a positive and significant effect on breakthrough innovation. A one-standard deviation increase in founder control is associated with an increase of approximately 1.164 breakthrough patents (6.262×0.186), an economically substantial effect equivalent to 82% of the sample mean patent count. In sum, these findings provide strong causal evidence that founder control fosters breakthrough innovation.

4.2.2. Heckman selection model

A second endogeneity concern is selection bias. A founder-led firm's innovative performance hinges on its founding team's characteristics, such as specific human capital and composition (Protojerou et al., 2017). These attributes are heterogeneous and often unobserved. This innate capability allows a firm to pursue ambitious, high-risk R&D projects that inherently carry a significant chance of yielding no patent. Our sample, however, includes only firms that successfully patent. It may therefore systematically exclude the capable but "unlucky" innovators whose ambitious projects failed. If this unobserved capability correlates with the potential for a breakthrough, our estimates will be biased.

To address this, we employ the Heckman two-stage estimation procedure. The credibility of this model hinges on a valid exclusion restriction, i.e., a variable that influences the selection decision (to patent or not) but does not directly influence the outcome variable (breakthrough innovation), other than through its effect on selection. For this, we use the average R&D intensity of the firm's industry in a given year (excluding the focal firm). The theoretical justification is twofold. First, higher industry-wide R&D activity can increase the pressure for a firm to patent its own discoveries to signal quality or protect intellectual property in a competitive environment. Thus, it plausibly affects the likelihood of a firm appearing in our patenting sample. Second, this variable should not have a direct effect on a focal firm's breakthrough innovation conditional on its own R&D spending and other firm characteristics. The success of a firm's innovation is driven by its internal resources and governance rather than by the patenting noise of its industry peers.

The untabulated first-stage probit model estimates the likelihood that a firm has at least one patent in a given year, using our identifying variable and the full set of controls. The second-stage outcome equation then includes the inverse Mills ratio calculated from the first stage. As reported in Column(3) of Table 4, the marginally significant inverse Mills ratio suggests the presence of selection bias. Crucially, after correcting for this bias, the coefficient on founder control remains positive and significant. This result provides strong assurance that our main finding is not a consequence of the sample limited to patenting firms.

4.2.3. Propensity score matching approach

A significant challenge to our causal claim is that the retention of founder control is an endogenous governance choice, not a random assignment. Firms that maintain founder control may systematically differ from those that do not along observable dimensions that also correlate with innovation. For instance, the presence of powerful institutional investors or venture capitalists often leads to a dilution of founder equity while imposing pressures for short-term, less risky projects (Aghion et al., 2013). Conversely, younger and smaller firms might feature high founder control and greater innovative agility simply due to their lifecycle stage. To disentangle the effect of founder control from these confounding observable characteristics, we employ propensity score matching (PSM).

Our procedure begins with estimating a probit model where the dependent variable is a dummy for founder control, defined by the >50 % equity threshold. This provides a theoretically grounded and empirically unambiguous "treatment" status essential for the matching procedure. The model includes the full set of covariates from our baseline specification to predict the propensity score for each firm-year. We then match each treated firm to a non-treated firm using a nearest-neighbor algorithm, imposing a strict caliper of 0.001 to ensure a high degree of similarity between matched pairs.

Table 5 presents the results, starting with the critical diagnostic tests that validate our matching procedure. Panel A displays the probit regression results before and after matching. For the pre-match specification in Column (1), numerous covariates are highly significant predictors of founder control, yielding a pseudo- R^2 of 0.485. This suggests that significant observable differences exist. In stark contrast, the post-match regression in Column (2) shows these covariates are now jointly insignificant, and the pseudo- R^2 plummets to 0.018. This dramatic reduction indicates that the matching has successfully mitigated the systematic variation explained by these observables. Panel B provides further corroboration, showing that the mean differences in all covariates between the treated and control groups are statistically insignificant post-match. Collectively, Panels A and B provide strong evidence that our matching procedure has created a well-balanced sample where the control group serves as a valid counterfactual.

Having established the quality of the match, we can now more confidently estimate the average treatment effect on the treated

Table 5

Endogeneity tests II: propensity score matching estimate.

Panel A: Pre-match propensity score regression and post-match diagnostic regression				
Dependent variable: Dummy equals 1 for founders with absolute control (>50), and 0 otherwise				
	Pre-match		Post-match	
	(1)		(2)	
<i>R&D</i>	0.007 (0.010)		−0.002 (0.017)	
<i>Size</i>	0.088* (0.049)		0.028 (0.080)	
<i>Age</i>	−0.222*** (0.051)		−0.006 (0.083)	
<i>Roa</i>	0.717 (0.547)		0.254 (0.883)	
<i>Current</i>	−0.007 (0.010)		−0.007 (0.016)	
<i>Lev</i>	−0.591* (0.307)		−0.009 (0.490)	
<i>Bm</i>	−0.855*** (0.275)		−0.233 (0.447)	
<i>Top1</i>	0.137*** (0.005)		−0.006 (0.008)	
<i>Balance</i>	1.588*** (0.075)		−0.107 (0.127)	
<i>Board</i>	0.424** (0.166)		−0.303 (0.278)	
<i>Inst</i>	−0.059*** (0.002)		−0.001 (0.003)	
<i>Tmtage</i>	−0.044*** (0.009)		0.012 (0.014)	
Firm FE	Yes		Yes	
Year FE	Yes		Yes	
Pseudo R^2	0.485		0.018	
N	5971		1093	

Panel B. Difference in firm characteristics				
	Treated	Controls	Difference	t statistic
<i>R&D</i>	17.071	17.173	−0.102	−0.584
<i>Size</i>	21.240	21.263	−0.023	−0.476
<i>Age</i>	1.132	1.166	−0.034	−0.782
<i>Roa</i>	0.058	0.059	−0.001	−0.262
<i>Current</i>	4.380	4.263	0.117	0.496
<i>Lev</i>	0.279	0.285	−0.005	−0.527
<i>Bm</i>	0.396	0.392	0.005	0.495
<i>Top1</i>	32.750	32.043	0.707	0.937
<i>Balance</i>	1.050	1.053	−0.003	−0.070
<i>Board</i>	2.199	2.188	0.011	1.186
<i>Inst</i>	21.747	19.731	2.015	1.583
<i>Tmtage</i>	47.064	47.283	−0.220	−1.182

Panel C. Propensity score matching estimate				
	Treated	Controls	Difference	t statistic
Unmatched	2.248	1.916	0.177	1.87
Matched	2.749	1.312	1.436	3.54

Note: Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

(ATT). Panel C reports this result. In the matched sample, founder-controlled firms produce significantly more breakthrough innovations than their observationally similar counterparts. Notably, the ATT in the matched sample is substantially larger than the simple difference in the unmatched sample. This suggests the presence of a negative selection bias: firms that cede founder control often do so for reasons (e.g., pressure from short-term oriented investors, lack of founder vision) that are themselves negatively correlated with the pursuit of breakthrough innovation. By creating a more precise counterfactual, PSM alleviates the bias on observables, revealing an effect that better approximates the "true" impact of founder control on breakthrough innovation.

4.2.4. Further tests on reverse causality

A final consideration is the threat of reverse causality. A compelling alternative explanation is that the prospect of future innovation influences founder control. For instance, founders possessing private information about their firm's breakthrough potential might

strategically retain or increase their equity stakes to maximize returns from future successes, a form of anticipatory behavior noted in the entrepreneurship exit literature (DeTienne et al., 2015). If true, our baseline estimates would capture an anticipation effect rather than the causal influence of control on innovation. While our use of lagged independent variables partially mitigates this, we conduct two additional analyses to address this concern.

First, if the anticipation effect were the primary driver, the observed relationship should be disproportionately influenced by "innovation superstars", the very firms where such foresight would be most consequential. We test this by sequentially excluding top innovators above the 99th, 95th, and 90th percentiles. The results in Column (1)–(3) of Table 6 contradict this prediction. The coefficient on our variable of interest remains positive, statistically significant, and remarkably stable across these increasingly restrictive specifications. This stability suggests our finding is a systemic pattern, not one skewed by a few extreme outliers.

Second, the anticipation effect implies the marginal effect of founder control should be significantly stronger at the upper tail of the innovation distribution. We examine this using quantile regression. The results in Columns (4)–(6) show that the positive and significant effect of founder control is not confined to the top quantiles but is present across the 25th, 50th, and 75th percentiles. The fact that founder control matters across the performance spectrum, rather than being concentrated among the highest performers, casts considerable doubt on this reverse causality narrative.

4.3. Robustness checks

4.3.1. Alternative measures and innovation horizons

We test the robustness of our main finding to alternative measures for both founder control and innovation outcomes, with the results presented in Table 7.

First, we test whether the innovation effect is driven by specific legally meaningful control thresholds rather than the linear specification of our baseline model. We specify two binary indicators based on a three-year average of founder equity to isolate control at the 51 % simple majority threshold (*FC51*) and the 34 % veto power threshold (*FC34*). The results in Columns (1) and (2) show that the innovation-enhancing effect of founder control is robust to this non-linear specification, indicating its presence at discrete, economically critical thresholds of corporate power.

Second, we test whether our finding is specific to an equity-based proxy for founder influence. We replace our ownership measure with a governance-based proxy, founder board representation (*BC*), following Gao and Jain (2012). This variable captures influence wielded through direct oversight. The positive and significant coefficient on *BC* in Column (3) demonstrates that the effect holds across different dimensions of founder power and is not solely a function of cash flow rights.

Third, we examine whether our result is sensitive to the specific thresholds used to define an innovative patent. We employ a more stringent definition (*Pat10*), identifying patents in the top decile by forward citations and the bottom decile by backward similarity. Column (4) shows our baseline findings are robust and not dependent on the specific cutoffs used in our primary analysis.

Finally, we test the sensitivity of our results to the temporal measurement of innovation. We first address the concern that our finding is specific to the $t + 3$ horizon, and Columns (5) and (6) confirm its robustness to alternative $t + 1$ and $t + 5$ lags. We then address the noise inherent in single-year data by using a cumulative $t + 1$ to $t + 3$ patent count to better capture the innovation cycle. The stability of the coefficient in Column (7) confirms our result reflects a sustained economic effect rather than a consequence of our measurement choice.

4.3.2. Alternative specification strategies

We next subject the baseline findings to a battery of alternative specifications, with the results reported in Table 8.

We begin with the most transparent, albeit restrictive, OLS regression in Column (1), using the subsample with positive patent counts. While this approach is vulnerable to selection bias by design, it serves as a crucial benchmark even when analysis is confined to patent-producing firms. The coefficient on founder control is positive and significant at the 5 % level, establishing a preliminary positive association and suggesting our main finding holds even in a simpler specification.

Table 6

Endogeneity tests III: reverse causality checks.

	Dependent variable: <i>Patents</i> _{$t+3$}					
	(1) Excluding top 1 %	(2) Excluding top 5 %	(3) Excluding top 10 %	(4) Q25	(5) Q50	(6) Q75
<i>FC</i>	0.642*** (0.164)	0.739** (0.327)	0.813*** (0.236)	2.534*** (0.417)	1.393*** (0.485)	1.797*** (0.594)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.350	0.330	0.284	–	–	–
N	5711	5237	4299	6092	6092	6092

Note: Columns (1)–(3) sequentially trim top innovators with breakthrough patent counts above the 99th, 95th, and 90th, respectively. Columns (4)–(6) report quantile regressions at the 25th, 50th, and 75th percentiles of the conditional breakthrough innovation distribution. Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 7

Robustness checks I: alternative measures and innovation horizons.

	Dependent variable						
	<i>Patents</i> _{t+3}	<i>Patents</i> _{t+3}	<i>Patents</i> _{t+3}	<i>Pat10</i> _{t+3}	<i>Patents</i> _{t+1}	<i>Patents</i> _{t+5}	<i>Patents</i> _{3yrs}
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<i>FC</i> ₅₁	0.190*** (0.038)						
<i>FC</i> ₃₄		0.237*** (0.038)					
<i>BC</i>			0.920* (0.551)				
<i>FC</i>				0.659*** (0.153)	0.377*** (0.131)	0.793* (0.434)	0.600*** (0.072)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo <i>R</i> ²	0.444	0.444	0.444	0.547	0.437	0.463	0.681
<i>N</i>	6092	6092	6092	6092	5718	4032	6092

Note: Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Table 8

Robustness checks II: alternative specifications.

	Dependent variable: <i>Patents</i> _{t+3}							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	OLS	Poisson	Negative Binominal	Zero-inflated	PPML	PPML	PPML	PPML
<i>FC</i>	1.134** (0.396)	0.709*** (0.152)	0.689*** (0.153)	0.588*** (0.180)	0.613*** (0.179)	0.726*** (0.192)	0.881*** (0.168)	0.687*** (0.236)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Baseline FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Additional FE	–	–	–	–	Ind-Year	City-Year	–	–
Interaction	–	–	–	–	–	–	Controls×Trend	Controls×City
Pseudo <i>R</i> ²	–	0.514	0.349	–	0.454	0.470	0.445	0.449
<i>N</i>	3597	7476	7476	7476	6067	5090	6092	6092

Note: Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Having established this benchmark, we turn to more appropriate count-data models. Columns (2) through (4) employ the Poisson, negative binomial, and zero-inflated Poisson estimators, respectively. These models accommodate different assumptions about the patent data-generating process, from equidispersion (Poisson) to overdispersion (negative binomial) and dual-state zero generation (zero-inflated). Across all three specifications, the coefficient on founder control remains positive and highly significant, suggesting that our main findings are robust to the specific distributional assumptions imposed on the count data.

Next, we address concerns about unobserved time-varying heterogeneity at the industry and city levels. Column (5) introduces industry-by-year fixed effects, and Column (6) includes city-by-year fixed effects. In both cases, the coefficient on *FC* remains stable and highly significant, suggesting the relationship is not confounded by temporal shocks specific to certain industries or locations.

Finally, we test the robustness of our finding to more flexible forms of heterogeneity by allowing the effects of control variables to vary. Column (7) includes interactions between the controls and a linear time trend, while Column (8) includes interactions between the controls with city dummies. Even within these demanding specifications, which absorb a vast amount of heterogeneity in the impact of firm characteristics, the coefficient on founder control remains positive and statistically significant. Collectively, these tests provide strong evidence for a robust relationship between founder control and breakthrough innovation.

4.3.3. Subsample analysis

We then investigate three contextual dimensions that theory suggests may condition the innovation-enhancing effect of founder control, with the results presented in Table 9.

First, we consider capital market tenure. Founders who maintain control at the IPO often accept a valuation discount, whose financial benefits typically materialize after approximately five years of public trading (Fattoum-Guedri et al., 2018). We therefore partition our sample by listing age relative to the industry-year median (4 years in our data). Column (1) shows no significant effect of founder control among newly listed firms, whereas Column (2) reports a positive and significant coefficient for more mature firms. This pattern indicates that the innovation premium from founder control is amplified over time. As a firm moves beyond the intense scrutiny of its early public years, it builds reputational capital with the market. This, in turn, eases financing constraints, allowing founder-controlled firms to fund the large-scale, high-risk ventures that generate breakthrough innovations.

Table 9

Robustness checks III: subsample analysis.

	Dependent variable: $Patents_{t+3}$						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	Newly listed firms	Mature firms	Innovation city	Non-innovation city	High competition	Medium competition	Low competition
FC	0.227 (0.282)	0.656*** (0.248)	0.787*** (0.295)	0.504 (0.656)	1.727** (0.792)	−0.133 (0.294)	0.783*** (0.299)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.421	0.485	0.455	0.406	0.419	0.459	0.454
N	3369	2261	3141	2949	1659	1766	1785

Note: Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

Second, we turn to geographic agglomeration. High-tech clusters provide rich networks of research institutions, venture capital, and specialized talent that can amplify a founder's innovative initiatives (Moretti, 2021). Using the Ministry of Science and Technology list of national innovation-city pilots, we identify firms headquartered in these clusters. Column (3) shows a large and significant estimate for firms in innovation cities, whereas Column (4) shows an insignificant estimate in non-innovation cities. Local ecosystem quality therefore appears to condition the effectiveness of founder-led governance in fostering breakthrough innovation.

Third, we analyze the role of product-market competition, proxied by the Lerner index. Economic theory posits an inverted-U relationship, where innovation thrives under weak or strong competition but stagnates at moderate levels (Aghion et al., 2005). We estimate the effect of founder control within low, medium, and high competition terciles. Consistent with this theory, Columns (5) and (7) show significant positive effects in both low- and high-competition settings. By contrast, Column (6) reports an insignificant effect in mid-competition markets, where firms possess neither the ample rents to fund risky projects nor the existential pressure to innovate to escape competition. Founder control thus fosters breakthrough innovation when firms face either intense competitive pressure or possess sufficient market power, but offers limited advantage when neither condition prevails.

4.3.4. Effect of founder shareholding reductions

To further substantiate our theory, we test a key corollary that the innovation-enhancing effect of founder control should be attenuated when founders signal a weaker long-term commitment by reducing their ownership stakes. We examine this by testing whether the innovation effect is weakened for firms where founders have recently divested part of their equity. We construct two measures to capture this dynamic over a rolling three-year window.¹ The first is a binary variable, $1(Divestiture)$, which equals 1 if the founding team's aggregate stake decreased by at least 5% during the period and 0 otherwise.² The second is a continuous variable, $Divestiture$, which quantifies the cumulative percentage decline in founder ownership over the same interval. We then interact each measure with FC in our baseline specification.

Fig. 2 provides graphical evidence consistent with this prediction. The plotted relationship between founder control and patent applications is substantially steeper for firms without recent ownership reductions (solid line), whereas the slope is nearly flat for firms where founders have reduced their stake (dashed line). This visual pattern indicates that the innovation-enhancing effect of founder control is markedly attenuated when founders divest.

Table 10 formally confirms this pattern. Using the binary indicator in Column (1) and the continuous measure in Column (2), we find both interaction terms are negative and statistically significant. These results provide strong support for the argument that founder control's efficacy is conditional on the founder's demonstrated commitment. While this analysis does not prove the logical contrapositive of our main hypothesis, it provides compelling evidence that sustained founder ownership is critical for preserving the long-horizon incentives that fosters breakthrough innovation.

5. Mechanism analysis

We now test the four mechanisms hypothesized to underlie the relationship between founder control and breakthrough innovation. We examine two internal mechanisms—talent investment (Hypothesis 2a) and internal discipline (Hypothesis 2b)—and two external mechanisms—strategic reorientation (Hypothesis 3a) and technological resilience (Hypothesis 3b).

¹ The three-year window is chosen primarily to align with the forward-looking nature of our dependent variable $Patents_{t+3}$, ensuring a consistent temporal structure in our research design. This timeframe also strikes a balance between shorter windows, which exhibit insufficient variation in ownership changes, and longer windows, which risk diluting the causal link between a divestment decision and subsequent innovation outcomes.

² Our choice of the 5% threshold is grounded in both institutional and methodological considerations. Institutionally, China's Securities Law mandates public disclosure for any ownership change of 5% or more, legally define it as a material event for market participants. Methodologically, this cutoff allows us to isolate a strong signal of waning founder commitment from the noise of minor technical adjustments.

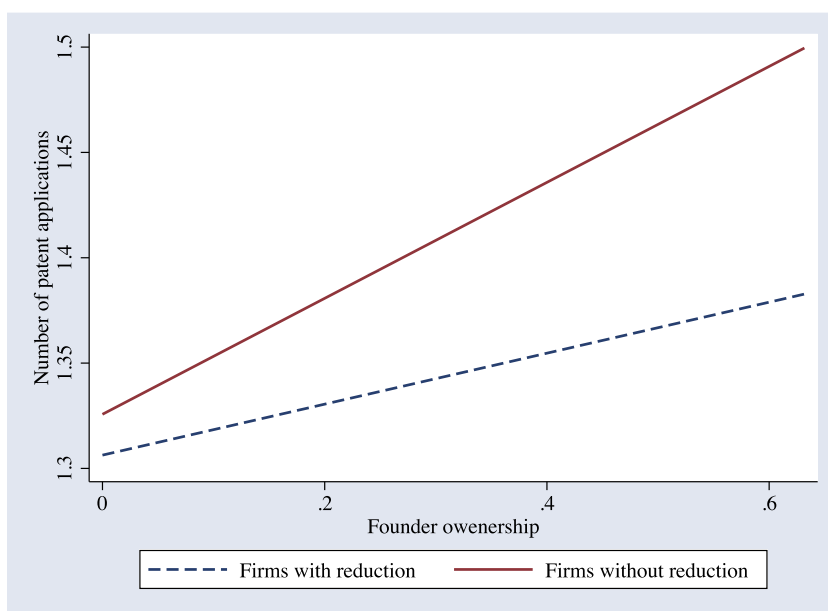


Fig. 2. Linear trend of founder control on innovation by shareholding reduction.

Table 10

The effect of shareholding reductions.

	Dependent variable: $Patents_{t+3}$	
	(1)	(2)
FC	0.931*** (0.145)	1.432*** (0.406)
$1(\text{Divestiture})$	0.162*** (0.037)	
$FC \times 1(\text{Divestiture})$	-0.510*** (0.181)	
$Divestiture$		-0.164 (0.523)
$FC \times Divestiture$		-1.663*** (0.563)
Controls	Yes	Yes
Firm FE	Yes	Yes
Year FE	Yes	Yes
Pseudo R^2	0.444	0.457
N	6092	1146

Note: Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

5.1. Intensive talent investment

We first test [Hypothesis 2a](#) that founders foster innovation by concentrating resources on elite talent. This requires a proxy that distinguishes the intensity of investment in human capital from the overall R&D scale. While metrics such as total R&D spending or headcount capture the scale of R&D, they fail to reveal the allocative pattern, namely whether to distribute resources broadly or to concentrate them on select individuals. We therefore employ R&D expenditure per R&D employee ($RDper$) as a proxy for the intensity of a firm's human capital strategy. A high ratio suggests an exploration-oriented strategy, reflecting a deliberate choice to endow R&D personnel with superior resources, such as advanced instrumentation, greater latitude for ambitious projects, or compensation sufficient to attract and retain exceptional individuals. Thus, this measure allows us to distinguish a commitment to nurturing high-potential innovation from the simpler goal of managing a large research function.

The results in Column (1) of [Table 11](#) support [Hypothesis 2a](#). The coefficient on the interaction term, $FC \times RDper$, is positive and statistically significant. It demonstrates that the positive effect of founder control on a firm's future patent output is significantly amplified in firms with higher per-capita R&D investment. The magnitude implies that a one-unit increase in $RDper$ increases the marginal effect of founder control on patenting by 0.833. This suggests that a key mechanism through which founder-controlled firms achieve breakthrough innovation is by directing resources to create a capital-intensive environment, enabling their technical talent to

Table 11
Mechanism analysis.

	Dependent variable: $Patents_{it+3}$			
	(1)	(2)	(3)	(4)
<i>FC</i>	0.884*** (0.312)	0.654*** (0.119)	0.420 (0.323)	0.427 (0.306)
<i>RDper</i>	−0.039 (0.060)			
<i>FC</i> × <i>RDper</i>	0.833*** (0.085)			
<i>Mngex</i>		0.267*** (0.079)		
<i>FC</i> × <i>Mngex</i>		−0.676*** (0.133)		
<i>Churn</i>			−0.125* (0.067)	
<i>FC</i> × <i>Churn</i>			0.394** (0.182)	
<i>Blocked</i>				0.180** (0.077)
<i>FC</i> × <i>Blocked</i>				0.544** (0.216)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.444	0.444	0.444	0.444
N	6041	6092	6092	6092

Note: Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

perform at the frontier of knowledge.

5.2. Internal discipline

We next examine [Hypothesis 2b](#), which posits that founders enhance innovation by imposing rigorous discipline. We proxy this with the firm's management expense ratio (*Mngex*). This measure directly captures bureaucratic slack and administrative costs that stand in contrast to value-creating activities. While some administrative expenditure is necessary, we argue that a higher ratio, on average, reflects the organizational slack and potential inefficiency that our theory suggests entrepreneurial founders actively minimize. This allows us to test whether the innovation effect of founder control is more pronounced in leaner firms.

The results in Column (2) support Hypothesis 2b. The interaction term *FC* × *Mngex* yields a negative and significant coefficient. This finding suggests that the innovation outcomes from founder control diminish as a firm's administrative overhead increases. The magnitude indicates that a one-unit increase in the management expense ratio reduces the marginal effect of founder control by 0.676. This evidence suggests that a crucial mechanism for founder-led innovation is the ability to maintain a disciplined focus on value creation, thereby preventing resources from being diluted by administrative inefficiency.

5.3. Strategic reorientation

We then turn to how founder-led firms interact with their external environment. We posit in [Hypothesis 3a](#) that founder control fosters innovation through strategic reorientation in the face of industry turbulence. To test this, we proxy for environmental turbulence using an industry churn rate, calculated as the number of deregistered firms divided by the number of active firms within a given province-industry-year. This measure captures a key element of the Schumpeterian process of creative destruction. High environmental churn creates both existential threats and growth opportunities. This measure goes beyond general economic fluctuations, which may affect all firms, to pinpoint industry-specific dislocations where strategic reorientation is essential for survival.

Column (3) of [Table 11](#) shows a positive and significant interaction term *FC* × *Churn*. This result is consistent with Hypothesis 3a, indicating that the relationship between founder control and innovation is stronger in more turbulent environments. The coefficient's magnitude suggests that the marginal effect of founder control on innovation increases by 0.394 for every one-unit increase in *Churn*. The finding suggests that founder control provides a critical advantage during industry shakeouts, enabling founder-led firms to capitalize on external shocks as opportunities to leapfrog competitors amidst the creative destruction within the market.

5.4. Technological resilience

Our final [Hypothesis 3b](#) posits that founder control enables firms to sustain their innovative trajectory and display technological self-reliance even when facing significant external adversity. Testing this mechanism requires operationalizing an environment where

such resilience would be critical. To this end, we construct a proxy for industry-level geopolitical pressure, *Blocked*. The proxy originates from the 1065 Chinese entities on the U.S. Bureau of Industry and Security's (BIS) Entity List.³ Using the Qichacha database, we manually assigned each entity to its primary industry, which allowed us to construct an annual industry-level measure of exposure to U.S. export controls. To create a clear distinction between environments of high and low adversity, we define *Blocked* as a dummy variable, equal to 1 if an industry's sanction count in a given year is above the sample median.

The results in Column (4) of Table 11 are consistent with Hypothesis 3b. The coefficient for the interaction term $FC \times Blocked$ is positive and statistically significant, with the marginal effect of founder control on innovation being 0.544 higher in industries facing above-median sanctions. This result demonstrates that the innovation premium associated with founder control is not only resilient to but is in fact magnified by external pressure. Converting adversity into a distinct innovative advantage appears to be a core capability of founder-led firms, driven by their unique governance and technological resolve.

6. Further analysis

Having established the main effect and its underlying mechanisms, we now explore key sources of heterogeneity that may condition the relationship between founder control and breakthrough innovation. We move from the structure of the founding team to its internal architecture, then to the attributes of the founders themselves, and finally to the temporal evolution of their influence. This multi-layered analysis extends beyond the question of *if* founder control matters to examine *how* its internal architecture and *for whom* its effectiveness is more pronounced. These findings support our theoretical shift from “who governs” to “how they govern”.

6.1. Single versus multiple founders: the role of founding team size

We begin by examining how the composition of the founding team—specifically, its size—shapes the efficacy of founder control. The literature suggests a fundamental trade-off (Wasserman, 2013). Larger teams bring greater cognitive diversity and broader human capital, yet they also introduce higher coordination costs and the potential for internal agency conflicts that can dilute shared belief. Conversely, single-founder firms benefit from a unified vision and rapid decision-making but face acute resource and knowledge constraints. In our sample, firms are almost evenly split between single-founder (50.8 %) and multi-founder origins, with the vast majority of the latter (41.1 % of the total sample) having teams of two to four members. This relatively balanced distribution provides a suitable setting for testing the proposed trade-off.

The results presented in Table 12 reveal a clear pattern. First, comparing Columns (1) and (2), we see the effect of founder control is positive and significant for both single-founder firms and multi-founder firms. However, the magnitude of coefficient for single-founder firms is substantially larger than the average for their multi-founder counterparts, suggesting the potent benefit of an unconflicted corporate strategy. This implies that, on average, the coordination costs within teams may dilute the innovative benefit of founder control.

To explore this further, we partition the multi-founder sample. The findings in Columns (3) and (4) uncover a sharp threshold. For firms founded by small teams (≤ 4 members), founder control has a strong positive effect on breakthrough innovation, an impact whose magnitude is statistically indistinguishable from that of a lone founder. In stark contrast, for firms with large founding teams (> 4 members), the effect becomes statistically insignificant and turns negative. This evidence suggests the governance advantage of multi-founder control is contingent. It thrives under a unified or tightly-knit command structure but is neutralized by the coordination costs and directional dissensus inherent in large founding teams. For the pursuit of breakthrough innovation, the unifying force of a singular guiding vision appears to be essential.

6.2. The internal architecture of founder control

The efficacy of founder control is not monolithic; it is contingent on the internal architecture (Botelho et al., 2025). To solve the fundamental challenge of translating a founder's *de jure* authority into *de facto* decisiveness, the governance architecture must resolve critical ambiguities regarding command (active or passive), hierarchy (who decides), and responsibility (who for what). We test these dimensions in turn, with the results presented in Table 13.

First, we examine the locus of command in single-founder firms, where the core challenge is closing the gap between the technological blueprint and operational execution. We partition the sample into founders serving only as board chair (*D.only*) and those holding the dual Chairman-CEO role (*Dual*). Column (1) shows that mere board oversight yields no significant innovation effect. In contrast, Column (2) reveals that combining formal authority with executive power generates a large and significant positive effect. This finding aligns with theories of strategic execution (Li et al., 2024), suggesting that for a lone founder, formal control is necessary but insufficient; it must be fused with hands-on executive leadership for decisive resource allocation.

Next, we turn to multi-founder teams, where the core challenge shifts from execution to coordination. To avert the decision-making gridlock that can plague egalitarian teams, a clear internal hierarchy is essential. We proxy for this hierarchy using two measures of equity concentration within the founding team: an inverse Herfindahl-Hirschman Index of founder stakes (*InvHHI*) and the team's equity concentration relative to its industry-year median (*RelCrm*). Columns (3) and (4) show a greater concentration of equity—and

³ Data source: <https://www.tclegal.cn/export-object>

Table 12

Further analysis I: the impact of founding team number.

	Dependent variable: $Patents_{t+3}$			
	(1)	(2)	(3)	(4)
	Single founder	Multiple founders		
		Full sample	≤ 4 founders	> 4 founders
<i>FC</i>	0.978** (0.466)	0.614*** (0.232)	0.996*** (0.280)	−0.427 (0.313)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.452	0.441	0.441	0.457
N	3297	2783	2042	741
Difference test (χ^2)	—	—	12.70***	

Note: Robust standard errors clustered at the firm level are reported in parenthesis. The reported χ^2 -statistic tests the equality of the *FC* coefficients in Columns (3) and (4), obtained from a seemingly unrelated regression (SUR) estimation. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

thus power—within the team is strongly associated with more breakthrough innovation. This finding refines the insights of [Hellmann and Wasserman \(2017\)](#), suggesting that while shared control is beneficial, a clear internal hierarchy, codified through asymmetric equity stakes, is crucial. It establishes a “first among equals,” mitigating coordination costs and enabling the team to act with the unified purpose required for high-risk projects.

Finally, we investigate the heterogeneity in founder roles ([Van Lancker et al., 2023](#)). For a lone founder, Column (5) shows that involvement in multiple business areas is positively associated with innovation, reflecting the necessity for a single leader to act as an integrator to compensate for the absence of partners. For multi-founder teams, the opposite logic prevails. Column (6) indicates that functional overlap among co-founders dampens innovation. Instead, Columns (7) and (8) show that these teams benefit from specialization, particularly when founders focus on core R&D roles. This pattern is consistent with a classic division-of-labor argument that founding teams create value not by replicating skills, but by deploying complementary expertise.

These results, taken together, reveal a powerful organizing principle for founder-led innovation. There is no single optimal design. Rather, the architecture must be tailored to the team’s initial composition. Single founders succeed through the consolidation of authority and knowledge. Multi-founder teams succeed through the differentiation of power and expertise. Ultimately, the ability of founder control to foster breakthrough innovation hinges on this deliberate internal engineering, providing a clear example of how the architecture of authority, rather than merely the presence of a founder, drives transformative outcomes.

6.3. Founder attributes as conditioning factors

While founder control provides the structural authority to pursue breakthrough innovation, the capacity to effectively execute may depend on the founder’s personal attributes. We propose that founder control and key founder attributes are complements. Control provides the power to act on distinctive insights, while attributes provide the insights themselves. We explore four such attributes: the founder’s demographic profile (age), formative experiences (overseas exposure or a background in a research institute), and positional role (serving as chief technology officer). The results are presented in [Table 14](#).

First, we consider founder age, which proxies for different forms of experiential capital. For single-founder firms, our analysis reveals a U-shaped pattern.⁴ Column (1) reveals that the innovation-enhancing effect of founder control is most pronounced for the youngest and the most senior founders relative to their middle-aged peers, a direct contrast to the well-documented inverted-U relationship in developed economies ([Azoulay et al., 2020](#)). This logic of individual human capital, however, is inverted in a team setting. Our untubulated analysis, using a continuous measure of the team’s average age, reveals a significant and negative interaction with founder control. Here, the benefits of individual experience appear to be overshadowed by the risks of collective rigidity and decision-making inertia, turning control into a potential impediment in aging leadership teams. These divergent paths suggest that youth may be a particularly powerful driver for founder-led breakthrough innovation.

Second, we examine the role of a founder’s international background as a critical formative experience. Exposure to foreign technological and market paradigms can broaden a founder’s cognitive horizons and provide access to novel knowledge ([Criaco et al., 2022](#)). We interact founder control with an indicator for whether any founder has prior overseas work or educational experience. Column (2) shows a large and significantly positive coefficient on the interaction term. This finding supports the view that international background acts as an enabling condition to overcome organizational parochialism. It empowers a leader to act decisively on

⁴ Our choice to test a non-linear relationship is theoretically grounded. Although standard human capital models often imply an inverted-U relationship, we posit that in a transitional economy, innovative potential may peak at both ends of the age spectrum. Youth brings the cognitive flexibility to defy legacy paradigms, while seniority provides the relational capital to navigate institutional complexity. The statistical insignificance of a simple linear interaction term for this subsample ($p > 0.10$) empirically validates this approach, indicating a linear model would obscure the underlying dynamic.

Table 13
Further analysis II: the impact of founding team structure.

	Dependent variable: $Patents_{t+3}$							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
	Locus of command		Internal hierarchy		Functional configuration			
	Single-founder firms		Multi-founder firms		Single-founder firms	Multi-founder firms		
	Chair only	Dual roles	Equity crn.	Relative crn.	Breadth	Overlap	R&D spec.	Non-R&D spec.
<i>FC</i>	1.016*	0.786	2.203***	0.703***	0.071	1.489***	1.398***	0.403
	(0.524)	(0.552)	(0.240)	(0.145)	(0.557)	(0.267)	(0.311)	(0.307)
<i>D_only</i>	0.120***							
	(0.033)							
<i>FC</i> × <i>D_only</i>	−0.171*							
	(0.088)							
<i>Dual</i>		−0.162***						
		(0.056)						
<i>FC</i> × <i>Dual</i>		0.273***						
		(0.064)						
<i>InvHHI</i>			0.829***					
			(0.176)					
<i>FC</i> × <i>InvHHI</i>			−1.655***					
			(0.378)					
<i>RelCrn</i>				−0.831*				
				(0.435)				
<i>FC</i> × <i>RelCrn</i>				1.884**				
				(0.882)				
<i>#BG</i>					0.023	0.008		
					(0.206)	(0.048)		
<i>FC</i> × <i>#BG</i>					0.532***	−0.303***		
					(0.141)	(0.058)		
<i>#BG_NRD</i>							0.055	
							(0.065)	
<i>FC</i> × <i>#BG_NRD</i>							−0.350***	
							(0.076)	
<i>#BG_RD</i>								−0.082***
								(0.031)
<i>FC</i> × <i>#BG_RD</i>								0.271***
								(0.047)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R^2	0.453	0.453	0.440	0.440	0.455	0.440	0.440	0.440
N	3289	3289	2781	2781	3264	2759	2756	2786

Note: The variables used in the interaction terms and regressions are defined as follows. In Column (1), *D_only* indicates whether the founder serves exclusively as the board chair without simultaneously holding a managerial role. Column (2) introduces *Dual*, which captures whether the founder combines the roles of both chairman and CEO. Column (3) uses *InvHHI*, the inverse Herfindahl-Hirschman Index of founder stakes, to measure the internal hierarchy within the founding team, while column (4) employs *RelCrn*, the relative equity concentration relative to the industry-year median, to provide an alternative measure. Column (5) and Column (6) feature *#BG*, a variable representing the number of unique professional backgrounds observed for each firm-year, encompassing functional areas such as production, R&D, finance, and legal, serving as a proxy for the breadth of team skills. Columns (7) and (8) disaggregate *#BG* into *#BG_NRD*, capturing the number of non-R&D professional backgrounds, and *#BG_RD*, the number of R&D-related backgrounds, respectively. Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

non-local knowledge, which a more consensus-driven governance structure might discount as unfamiliar or excessively risky.

Third, we delve deeper into the nature of a founder's professional background by focusing on prior experience in a research institute. We argue that such a background imparts more than merely scientific knowledge. It cultivates a “translational mindset”, a specific discipline for converting abstract discovery into viable technology (Jain et al., 2009). This training equips founders to forge a path from prototype to product. We interact founder control with an indicator for this experience. The positive and significant interaction term in Column (3) supports this argument. This suggests that control is most potent when wielded by founders whose professional history has already trained them in the art of converting scientific potential into technological reality.⁵

Finally, we test the synergy between founder control and technical leadership. The rationale is that founders in such roles possess

⁵ In an unabridged analysis, we also tested for the effect of prior experience at a university. We found the interaction term to be insignificant. This null result is consistent with our theoretical argument that the mission-driven and applied-science orientation of research institutes provides a more direct complement to founder control for generating breakthrough commercial innovation than the basic-research orientation often found in university settings.

Table 14

Further analysis III: the impact of founder attributes.

	Dependent variable: $Patents_{t+3}$			
	(1)	(2)	(3)	(4)
<i>FC</i>	0.392 (0.372)	0.693*** (0.161)	0.621*** (0.203)	0.722*** (0.140)
<i>Younger</i>	0.085** (0.034)			
<i>Senior</i>	−0.009 (0.077)			
<i>FC</i> × <i>Younger</i>	0.476** (0.198)			
<i>FC</i> × <i>Senior</i>	0.630*** (0.193)			
<i>Overseas</i>		−0.166*** (0.023)		
<i>FC</i> × <i>Overseas</i>		0.587*** (0.121)		
<i>Institute</i>			−0.179 (0.147)	
<i>FC</i> × <i>Institute</i>			0.865** (0.370)	
<i>Techposition</i>				−0.356** (0.171)
<i>FC</i> × <i>Techposition</i>				0.941*** (0.191)
Controls	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Pseudo R^2	0.454	0.444	0.444	0.444
N	3289	6092	6092	6092

Note: Column (1) analyzes the single-founder firm subsample. *Younger* and *Senior* are binary variables, taking the value of 1 if the founder's age falls within the first or third tercile, respectively, relative to their industry-year peers. The middle-aged group serves as the baseline. In Column (2), *Overseas* is a binary variable equal to 1 if any member of the founding team has overseas experience in either education or work. In Column (3), *Institute* is a binary variable equal to 1 if any member has prior experience in a research institute. In Column (4), *Techposition* is a binary variable equal to 1 if any member holds the position of chief technology officer. Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

both deep technological domain knowledge and the direct authority to shape innovation strategy. This co-location of expertise and authority is designed to minimize information loss between the laboratory and the boardroom, empowering the firm to act on deep technical insights without the delays of internal translation (Gu et al., 2024). We interact founder control with an indicator for a founder concurrently serving as the firm's chief technology officer (CTO). The positive and significant interaction term in Column (4) strongly affirms this logic. It demonstrates that founder control is more effective when it is institutionally fused with the firm's core technological decision-making apparatus.

Taken together, these findings suggest the value of founder control is realized when it is complemented with a founder's distinctive endowments—be it demographic, experiential, or positional. It is the instrument through which a founder's unique, and often non-verifiable, personal insights are forged into corporate strategy.

6.4. Governance window: the non-linear effect of founder tenure

A final question is whether the innovation premium from founder leadership is perpetual or finite. We focus on the tenure of founder-CEOs, where vision and executive authority are fused. While this fusion initially confers a decisive advantage, prolonged tenure can transform a founder from a farsighted asset into a source of organizational rigidity and cognitive entrenchment (Souder et al., 2012). A once-novel perspective thus risks hardening into dogma, creating a finite “governance window” for peak effectiveness. We test for this non-linear effect and, crucially, how it is conditioned by the founder-CEO's retention of ultimate control.

A preliminary visualization in Fig. 3 plots breakthrough innovation against founder tenure, suggesting a clear inverted U-shaped relationship. We formalize this observation in Table 15. Using a quadratic specification and applying the stringent testing criteria of Haans et al. (2016), Column (1) shows a statistically significant inverted U-shaped relationship for the full sample of founder-CEOs, with their innovative impact reaching an apex at approximately 6.7 years before declining.

This aggregate “best-before” lifecycle, however, masks a fundamental heterogeneity rooted in power. The core insight emerges when we distinguish between founder-CEOs who remain the firm's actual controller and those who have ceded ultimate authority. Columns (2) and (3) reveal a sharp contrast. For founder-CEOs who retain control, the apex of their innovative contribution arrives at nearly 13 years. In sharp contrast, for those without ultimate control, the peak occurs in under 4 years, followed by a more pronounced decline.

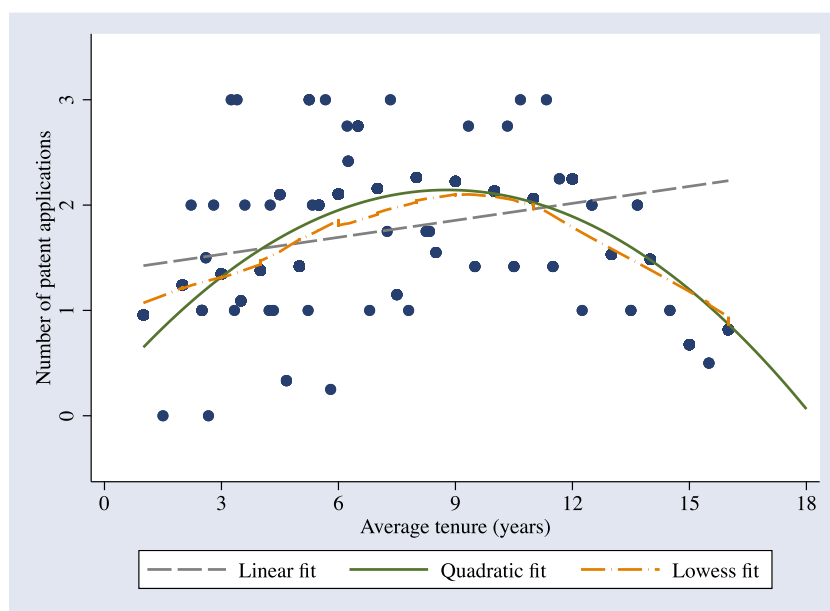


Fig. 3. Founder tenure and breakthrough innovation.

Table 15

Further analysis IV: the nonlinear effects of founder tenure.

	Dependent variable: $Patents_{t+3}$		
	(1) Full sample	(2) Actual controller	(3) Non-actual controller
<i>Tenure</i>	0.115** (0.052)	0.089** (0.035)	0.207*** (0.037)
<i>Tenure</i> ²	−0.009** (0.003)	−0.011*** (0.001)	−0.008** (0.004)
Controls	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Pseudo R^2	0.443	0.468	0.438
N	3123	1517	1541
Extreme point	6.673	12.826	3.964
Slope of lower bound [<i>t</i> statistic]	0.115** [2.217]	0.206*** [6.393]	0.088*** [2.554]
Slope of upper bound [<i>t</i> statistic]	−0.247*** [−2.543]	−0.131*** [3.789]	−0.381*** [−12.634]
Test for inverted U-shape (<i>p</i> value)	0.013	0.000	0.005

Note: The results are based on the following quadratic model:

$$\mathbb{E}(Patents_{it+3} | \cdot) = \exp(\alpha + \beta_1 Tenure_{it} + \beta_2 Tenure_{it}^2 + \mathbf{X}_{it}'\gamma + \eta_i + \delta_t) \quad (7)$$

where control variables ($\mathbf{X}_{it}$), firm fixed effects (η_i), and year fixed effects (δ_t) is identical to our baseline specification. The inverted U-shaped relationship is tested following the methodological guidance of Haans et al. (2016). Specifically, the significance of the quadratic term β_2 is evaluated, and the slopes at both the lower and upper bounds of the observed data range are calculated to confirm the presence of a turning point. Additionally, the test for an inverted U-shape was conducted by verifying whether the slope at the lower bound is significantly positive and the slope at the upper bound is significantly negative. Robust standard errors clustered at the firm level are reported in parenthesis. ***, **, and * indicate significance at the 1 %, 5 %, and 10 % levels, respectively.

This finding offers a substantive refinement to theories of founder imprinting and organizational inertia. The classic view posits that founders create a durable, often rigid, initial imprint on a firm's strategy and culture (Terbeck et al., 2022). Our results present a more dynamic, governance-contingent perspective. A founder's influence is not a static imprint but an evolving capability whose effective lifespan is endogenous to the firm's power structure. For a founder-CEOs without ultimate authority, their lingering presence may create goal ambiguity, transitioning from an asset to a liability. Conversely, sustained control appears to provide both the psychological security and the foundational impetus for a founder to co-evolve with the firm, continuously renewing their human capital. The central long-term challenge for founder-led firms, therefore, is not merely the maintenance of control, but the active management of

the leader's own lifecycle to ensure they remain the engine, and not the anchor, of the firm's adaptive capacity.

7. Conclusion

Using a 2008–2022 panel of Chinese high-tech firms, this study examines the effect of founder control on breakthrough innovation. We find a positive relationship, where founder control measured as equity held by active founders predicts a greater output of breakthrough innovation, which is identified using SBERT-based patent text analysis. The baseline positive association is identified as causal through a comprehensive identification strategy that includes an instrumental variable approach, sample selection models, matching estimators, and tests for reverse causality. We further explore the underlying mechanisms, including intensive talent investment, rigorous internal discipline, strategic reorientation amid industry turbulence, and technological resilience to geopolitical sanctions. Moreover, we document significant heterogeneity in this effect across key founding team characteristics, including its size, internal structure, attributes, and tenure. Taken together, our findings provide a comprehensive account of not only how this effect operates but also when it is most pronounced.

Our study's primary contribution is thus to reframe the debate on founder-led innovation. By moving beyond founder identity and focusing on founder control as a structural configuration, we provide a more robust framework for understanding how governance shapes a firm's capacity for exploration. Our findings suggest that founder control is not merely about a founder's personal trait; it is about an architecture of authority that insulates the firm from myopic pressures, enables patient capital deployment, and empowers belief-based commitments. This structural view provides a more comprehensive understanding of when and why founder-led governance can foster breakthrough innovation.

Our findings should be interpreted with two caveats. First, the majority of our sample predates the widespread adoption of DCS structure in China. The reliance on indirect evidence therefore necessitates caution in extrapolating our results to new governance regimes. Second, our study focuses on innovation as the primary outcome and does not analyze the potential trade-offs, such as risks of minority shareholder expropriation, that may accompany concentrated founder control.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A

As discussed in Section 5, Hypothesis 2a and Hypothesis 2b posit that founder control influences the firm's R&D resource allocation and internal discipline, respectively. Table A.1 provides evidence for this prerequisite link by testing whether founder control (*FC*) has a direct effect on our two observable mediating variables: intensive talent investment (*RDperEmp*) and internal discipline (*Mngex*).

The results lend support to our hypotheses. Column (1) shows that founder control is positively and significantly associated with future R&D expenditure per employee. Column (2) shows a significant negative association with the future management expense ratio. These findings provide direct support for the first stage of our proposed mediation mechanisms, strengthening the interpretation that founder control fosters breakthrough innovation by enabling these specific allocative choices.

Table A.1

The direct effect of founder control on observable mediators.

	(1)	(2)
	<i>RDper</i> (t+1)	<i>Mngex</i> (t + 1)
<i>FC</i>	0.849** (0.357)	−0.094** (0.022)
Controls	Yes	Yes
Baseline FE	Yes	Yes
Adjusted <i>R</i> ²	0.739	0.557
<i>N</i>	6041	6092

Note: Robust standard errors clustered at the firm level are reported in parenthesis. ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively.

Data availability

Data will be made available on request.

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